

HTTP Request and Response Headers

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1. Objective

Objective of this lab is to understand the HTTP Request and Response Headers.

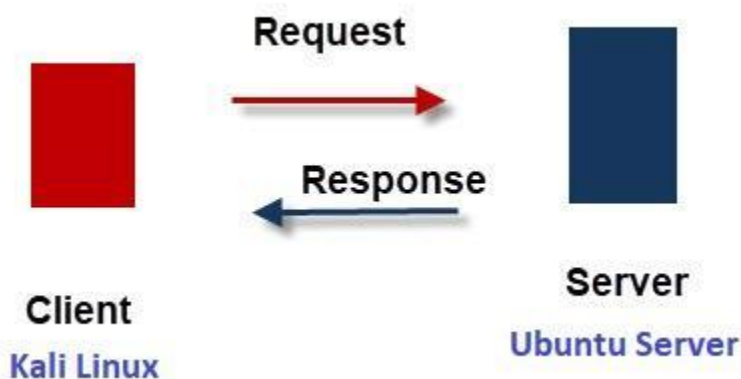
2. Prerequisites

Prerequisites	Specification
Source	Kali Linux
Target	Ubuntu Server(20.04, 64 bit) hosting the OWASP Insecure Web application

3. Problem Statement

The target application's web server may be vulnerable due to older version or unpatched version and so it's required to identify on which version of web server the target application is hosted on the web server.

4. Lab Diagram

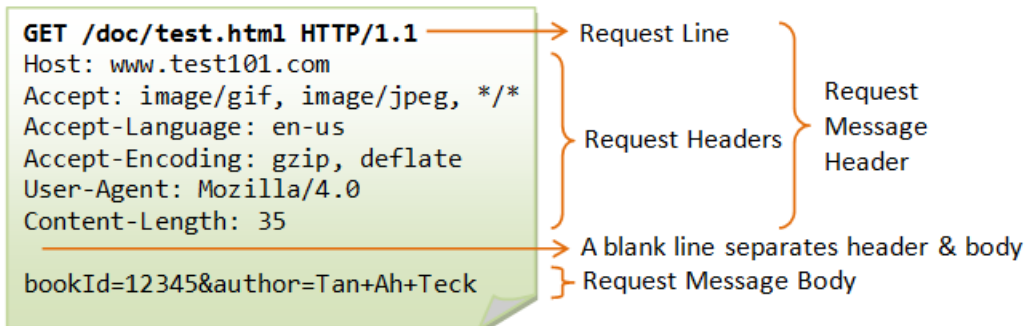


5. Summary

- All web browsers and web servers communicate with each other by sending HTTP requests and responses.
- A web browser communicates with a web server by sending an HTTP request for each requested resource (like HTML document, image, CSS etc).
- Request Header is present when you make a request to the server and the response header is present when the server sends a response back to the client/browser.

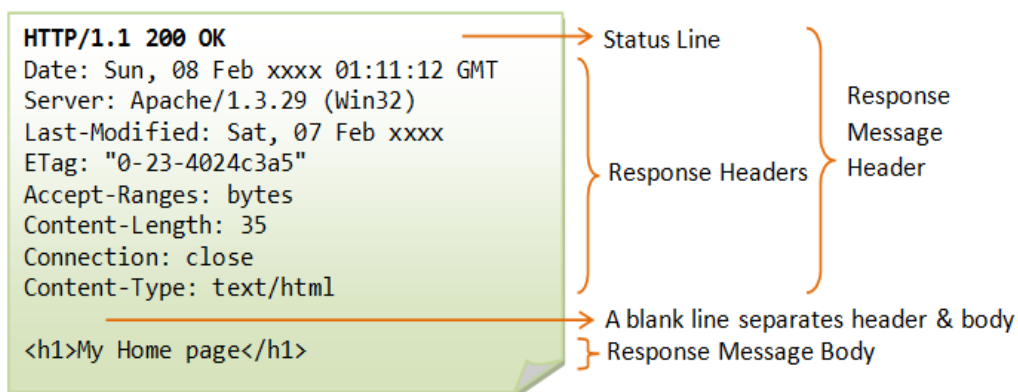
- **Request Header** contains information about the request such as the URL that you have requested, the method (GET, POST, HEAD), the browser used to generate the request and other info.

Example -



- **Response Header** is received by the browser from the server after the user sends a request for a particular page or resource and it contains information such as the encoding used in the content, the server software that is used on the server machine to generate the response and other information.

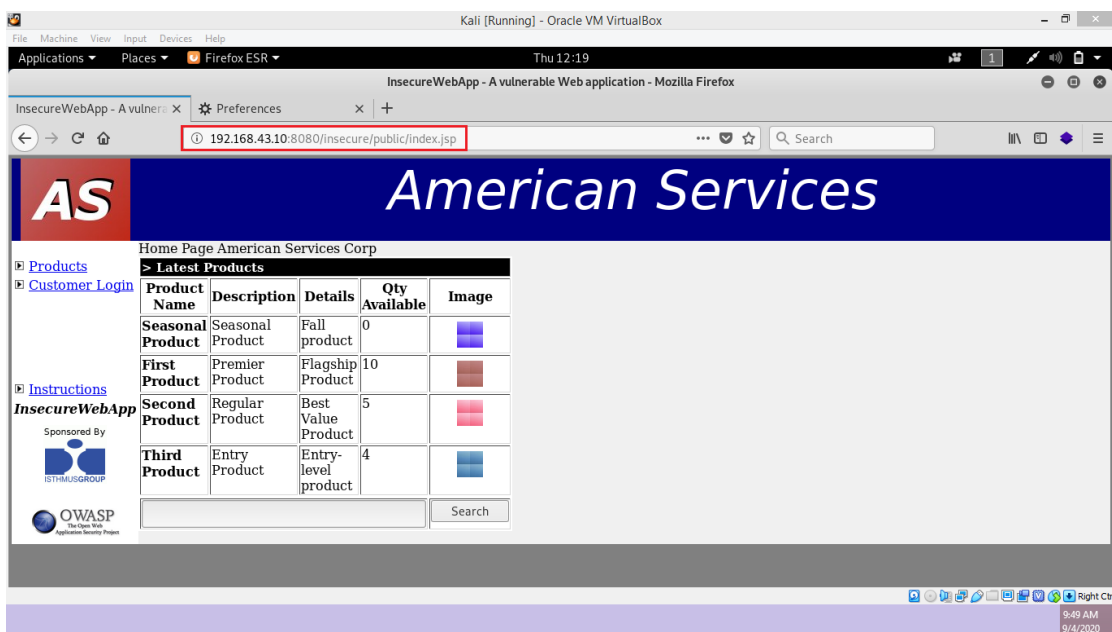
Example -



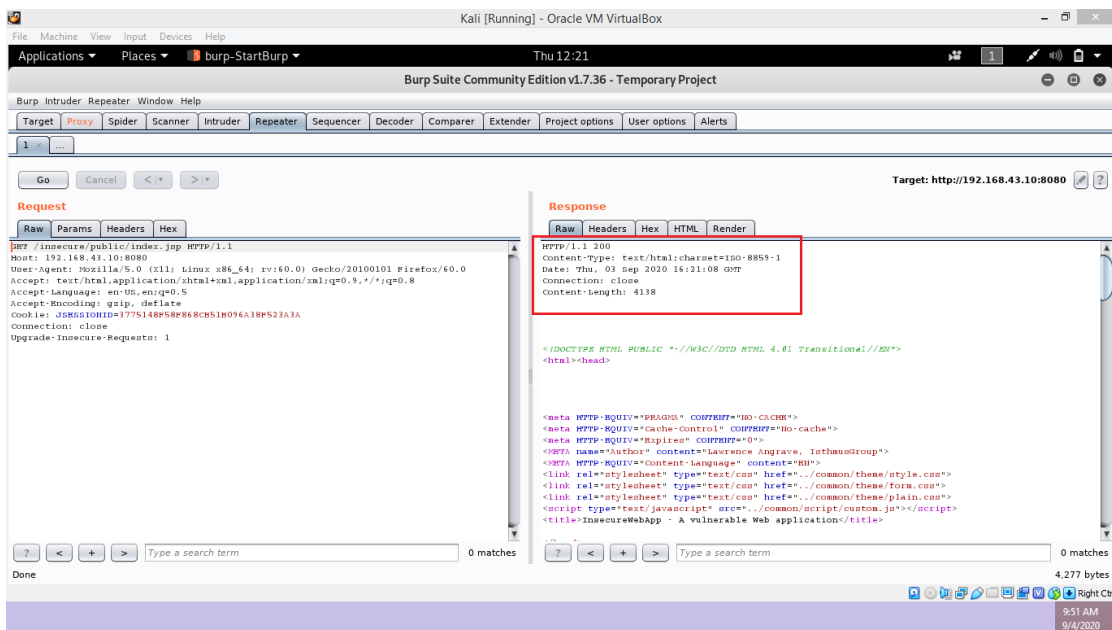
Most of the sites usually hide their server information in order to make it hard for hackers to know which software is being used on the server.

6. Solution

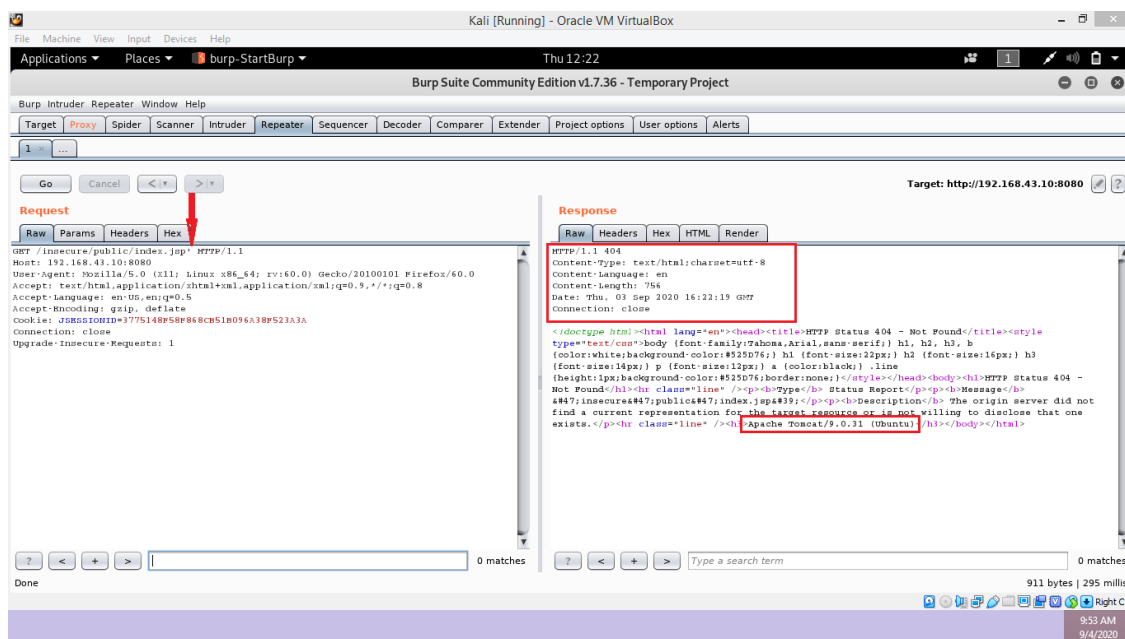
Step 1: Accessed the application which is hosted on the URL - <http://172.17.0.3:8080/insecure>.



Step 2: Intercepted the request in burp suite and sent to repeater as shown in screenshots.



Step 3: Here we will try to fetch sensitive information related to applications like web server version or operating system. Tampered the request and sent it to the server. It was observed that the web server version disclosed as "Apache tomcat/9.0.31" and operating system disclosed as "ubuntu" in response body. Using this information we can find out associated vulnerabilities with it.



7. References

- <http://www.steves-internet-guide.com/http-basics/>
- https://www.ntu.edu.sg/home/ehchua/programming/webprogramming/HTTP_Basics.html
- <https://www.ionos.com/digitalguide/hosting/technical-matters/http-header/>